

9.5 Cavalieri's Principle

Lesson Introduction

 Benchmark	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.		 Learning Target	I can use Cavalieri's principle to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms, and cones.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula. MA.7.GR.2.3 Solve mathematical and real-world problems involving volume of right circular cylinders. MA.912.GR.4.1 Identify the shapes of two-dimensional cross sections of three-dimensional figures.		N/A	
 Essential Questions	How can Cavalieri's principle be used to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms, and cones?			
 Lesson Narrative	Students are guided through the purpose of Cavalieri's principle and how it can be used to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms, and cones. The 9.5.2 Collaboration gives students several scenarios for the development of Cavalieri's principle while the rest of the lesson gives students the opportunity to put Cavalieri's principle into practice. This lesson also includes instruction on vertical and slant height.			
 Component Summary	9.5.1 Warm-Up (~5 min.)	Notice and wonder about records that are stacked differently (<i>SE p. 76</i>)	9.5.4 Guided Instruction (5-10 min.)	Defining slant height and vertical height (<i>SE p. 79</i>)
	9.5.2 Collaboration (5-10 min.)	Exploring scenarios to develop Cavalieri's principle (<i>SE p. 77</i>)	9.5.5 Guided Practice (10 min.)	Finding volume using the idea of Cavalieri's principle (<i>SE p. 80</i>)
	9.5.3 Guided Instruction (5-10 min)	Formalizing Cavalieri's principle (<i>SE p. 78</i>)	9.5.6 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 82</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Cavalieri's Principle for Three-Dimensional Solids - Slant Height - Vertical Height - Altitude <u>Additional Terms:</u> N/A		(Optional) Tangible Objects for 9.5.5 Guided Practice (i.e. stack of sticky notes, stack of chocolate squares, stack of compact discs, and glass vase)	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	On question 4b in 9.5.5 Guided Practice, students may forget to subtract parts of the vase. See note in teacher guide.	Consider having tangible visuals for students to access throughout this lesson to help fully solidify the concept of Cavalieri's principle.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder There are multiple chances for students to notice and wonder throughout the lesson. By having real-life scenarios, students can notice that if three-dimensional objects have the same areas of bases and the same heights, then the volumes are congruent. Give students the opportunity to notice and wonder in the lesson.	MA.K12.MTR.4.1: Discussion and Reflection Lesson 5 gives ample opportunities for students to discuss and construct their own knowledge. Provide structures and the environment that is conducive for students to engage in discussion of mathematics. Allow students to construct their own arguments with respect to Cavalieri's principle and the volume of a three-dimensional object.
 Differentiate Instruction	Consider providing objects for students to manipulate to show that if the heights of two three-dimensional objects are congruent and the areas of the bases are congruent (even if the shape of the bases are not congruent), then the volumes are congruent. There are several examples throughout the lesson to provide a visual. Teachers may also provide their own examples of visuals to demonstrate Cavalieri's principle.	
Lesson Notes:		

9.5.1 Warm-Up: Notice and Wonder

Component Overview: Students examine four pictures and record a Notice and Wonder.



9.5.1 Warm-Up: Notice and Wonder

Four different perspectives of the same stack of records are shown.



Provide students with the opportunity to share their noticings and wonderings with the class. By having real-life scenarios, students can notice that if three-dimensional objects have the same areas of bases and the same heights, then the volumes are congruent.

1. What do you notice?
Answers will vary.
All are viewed from different angles.
Some perspectives make it seem like there are more or less records that are in the stack.
2. What do you wonder?
Answers will vary.
Why are some stacked neatly and others not?
Why does a change in perspective make it seem like there are more or less records in the stack?

9.5.2 Collaboration: Exploring Cavalieri's Principle

Component Overview: Students collaborate and are given three distinct scenarios. Scenario 1 asks students to determine the relationship between the volumes of two stacks of pennies. Scenario 2 gives students two towers with the same heights and area but different bases. The last scenario describes the volumes between a cone and a pyramid.



9.5.2 Collaboration: Exploring Cavalieri's Principle

Scenario 1

Consider the two stacks of pennies, where each stack contains the same number of pennies. Discuss with your partner the relationship between the volumes of each stack.

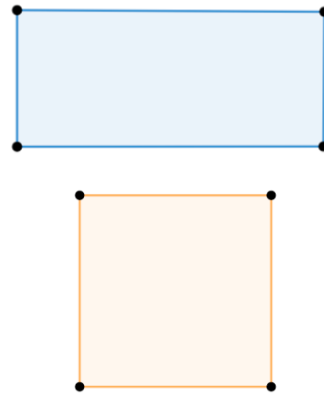


1. Summarize your discussion in the table below and have your partner initial that your summary is correct.

Summary	Partner's Initials
<i>Answers will vary.</i> <i>Volumes will be the same because the number of pennies is the same.</i>	

Scenario 2

Suppose two partners are both constructing towers that are to be 5 inches high. Partner A decided to build a tower using rectangles with areas of 16 square inches and partner B decided to construct a tower using squares with areas of 16 square inches. The base of each partner's tower is shown.



Discussion provides additional auditory entry-points. Consider utilizing sentence stems and providing ample time for students to grapple with academic language:

- "I think the volume of each of the stacks of pennies is _____, because the height of each stack of pennies is _____."
- "I notice in the picture that _____, which makes me think _____."



If necessary, provide students with a tangible stack of pennies for students to see the volumes are the same because the heights are the same.

For Scenario 2, consider using a graphing utility that has 3D capabilities.

9.5.2 Collaboration: Exploring Cavalieri's Principle (continued)

2. Discuss with your partner how the volumes will compare. Summarize your findings in the box below and have your partner initial that your summary is correct.

Summary	Partner's Initials
<i>Answers will vary. The volumes will be the same because they each have a height of 5 inches and the area of the bases will be the same.</i>	

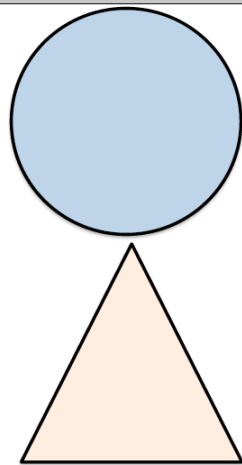


Ask the following questions to guide students through 9.5.2 Collaboration:

- How will the volumes be different if the areas of the bases and the heights are the same?
- Does the shape of the base affect the total volume?

Scenario 3

ABC ice cream shop decided to run a special for the week and they're offering two serving options for you to choose from. Option 1 is a cone with a circular base that has an area of 4π square inches and a height of 4 inches. Option 2 is a triangular pyramid container with a base of 4π square inches and a height of 4 inches. The base of each option is shown.



Ask the following questions to guide students through Scenario 3:

- What do you notice about the area of both bases?
- What do you notice about the heights of the cone and the pyramid?

3. Discuss with your partner how the volumes will compare to one another. Summarize your findings in the box below and have your partner initial that your summary is correct.

Summary	Partner's Initials
<i>Answers will vary. The volumes will be the same because the area of the bases will be the same.</i>	

9.5.3 Guided Instruction: Cavalieri's Principle

Component Overview: Students work through two questions on using Cavalieri's principle. The first question requires students to compare two three-dimensional objects and determine whether the volumes are equal. The second question ties in cross sections and formulas for area and volume.



9.5.3 Guided Instruction: Cavalieri's Principle



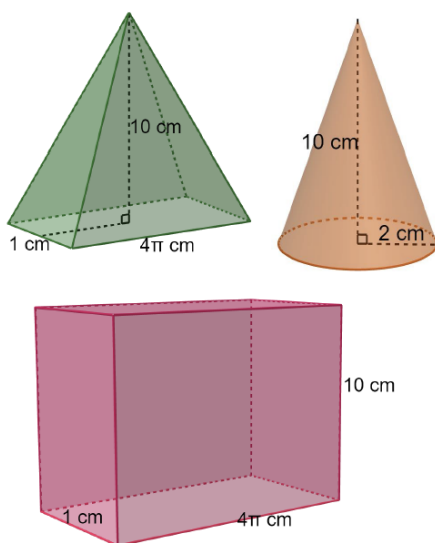
Cavalieri's Principle for Three-Dimensional Solids

If every plane parallel to the two bases of two solids results in cross sections of equal area and the two solids have congruent altitudes, then the solids have equal volumes.



What information do you need to know to determine whether the solids have the same volume?

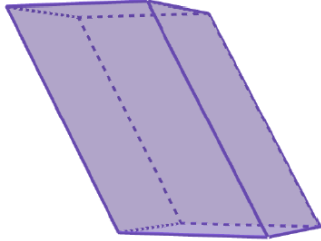
- Using Cavalieri's Principle, determine whether or not the following solids have the same volumes.



No. A cross section that is 1 cm from the base of the pyramid will not have an area that is the same as a cross section that is 1 cm from the base of the rectangle.

9.5.3 Guided Instruction: Cavalieri's Principle (continued)

2. An oblique prism is shown.



How do you determine the shape of the cross section of the oblique prism shown?

- a. What shape is each cross section of the oblique prism?

Parallelogram

- b. What is the area formula for the base?

$A = lw$

- c. What is the volume formula for an oblique rectangular prism?

$V = lwh$

- d. Discuss with your partner how the formula for the oblique rectangular prism compares to the formula for a right rectangular prism. Explain your reasoning.

Answers may vary. The formulas are the same just based off if the shape is two dimensional or three dimensional. The shape is designed different than normal.



Allow students the opportunity to share out their responses to question 2d. Use this as an opportunity to address misconceptions that arise during 9.5.3 Guided Instruction.

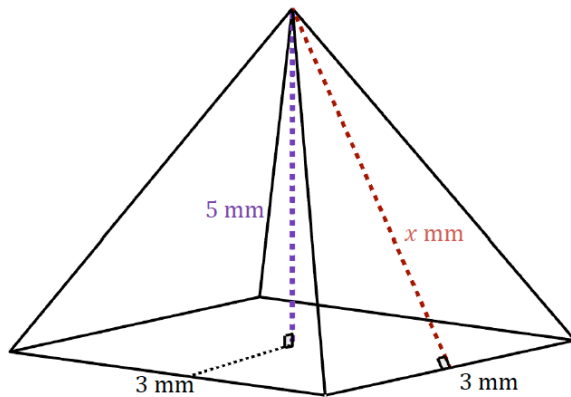
9.5.4 Guided Instruction: Slant Height vs. Vertical Height

Component Overview: Students are guided through a comparison of slant height and vertical height and how to find the slant height when given the vertical height.



9.5.4 Guided Instruction: Slant Height vs. Vertical Height

A square pyramid is shown with measurements in millimeters (mm).



Samantha and Julian each found the measure of the slant height, x , by first drawing a diagram and then applying the Pythagorean Theorem. Their work is shown.

Samantha	Julian
$c^2 = 3^2 + 5^2$	$c^2 = 1.5^2 + 5^2$



Have students discuss the similarities and differences between vertical height and slant height.



Why is slant height needed instead of vertical height to find the volume of a square pyramid?

9.5.4 Guided Instruction: Slant Height vs. Vertical Height (continued)

1. Discuss with your partner which student correctly set up their work to determine the slant height. Summarize your discussion.

Julian is correct because he used the correct distance from the base to the center for that leg side of the Pythagorean theorem.



The **slant height** of an object is the distance from the base to the apex along a lateral face, whereas the **vertical height or altitude** of an object is the perpendicular distance from the apex to the base.

2. Discuss with your partner which height is important when finding the volume of an oblique figure. Summarize your discussion.

Answers may vary. The vertical height or altitude is more important when finding the volume of an oblique figure.



Have a few students share their responses with other partners or classmates from question 1 and then repeat again in question 2.



Slant height is often a concept that students confuse with the traditional vertical height of a triangle. Spend some time discussing with students the impact of choosing the wrong value and why the values can be different.

9.5.5 Guided Practice: Cavalieri's Principle in the Real World

Component Overview: Students complete several questions relating to Cavalieri's principle. Question 1 discusses a stack of sticky notes. Question 2 gives students a picture of two different stacks of chocolate squares. Question 3 gives students a scenario with compact disks (CDs), and Question 4 gives students a picture of a vase and a context.



9.5.5 Guided Practice: Cavalieri's Principle in the Real World

1. Consider the stack of sticky note pads below.



If each pad is a square with side lengths of 3 inches and a height of 0.97 inches, approximate the volume of a stack of 8 sticky note pads.

$$V = (3)(3)(0.97)$$

$$V = 8.73$$

$$V = 8.73(8)$$

$$V = 69.84 \text{ in.}^3$$

2. Two stacks of 12 chocolate squares are shown.



Each chocolate square has the dimensions $1\frac{5}{8}$ inches (in.) by $1\frac{5}{8}$ in. Each square is $\frac{1}{8}$ in. thick. Find the volume of each stack.

Stack A	Stack B
$V = \left(1\frac{5}{8}\right)\left(1\frac{5}{8}\right)(1.5) = 3.96 \text{ in.}^3$	$V = \left(1\frac{5}{8}\right)\left(1\frac{5}{8}\right)(1.5) = 3.96 \text{ in.}^3$



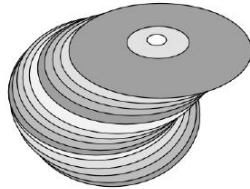
Does the volume change since the stack of sticky notes are not perfectly stacked? Why or why not?



What is something that you notice about the two stacks?
What is something you wonder about them?

9.5.5 Guided Practice: Cavalieri's Principle in the Real World (continued)

3. While cleaning out an attic, you found an old stack of compact discs (CDs) that were used to store music.



If each disc has an area of $36\pi \text{ cm}^2$, explain how you can find the approximate volume of a stack of 25 CDs.

Multiply the area of the disc by the total vertical height of the stack of CDs.

4. A zigzag crystal vase is shown, where the vase has a height of 10 inches (in.). The cross-sections that are parallel to the base are always rectangles that are 6 in. wide by 3 in. long.



- a. Determine the volume of the vase, assuming the crystal itself has no thickness.

$$V = lwh$$

$$V = (6)(3)(10)$$

$$V = 180 \text{ in.}^3$$



Consider having tangible visuals for these items as students are working through 9.5.5 Guided Practice. If available, try to bring some blank CDs in for students to have a visualization for question 3 to better connect to the content as they complete the question.

9.5.5 Guided Practice: Cavalieri's Principle in the Real World (continued)

- b. The crystal is actually half an inch thick on each of the sides and on the bottom. Approximately how much space is contained within the vase? Explain or show your reasoning.

$$6 - 1 = 5$$

$$3 - 1 = 2$$

$$10 - \frac{1}{2} = 9\frac{1}{2}$$

$$V = 5(2)\left(9\frac{1}{2}\right)$$

$$V = 95\text{in.}^3$$

Since each of the sides are 0.5 inches thick, the length and width of the cross-sectional rectangle will each decrease by 1 inch. Meanwhile, height of the vase itself will decrease by 0.5 inch due to the thickness of the glass.



Help students through question 4b if necessary. Some students may forget to subtract the $\frac{1}{2}$ inch thickness on each side of the vase, thus resulting in incorrect calculations. Also, students will only subtract $\frac{1}{2}$ inch on the bottom.

9.5.6 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



9.5.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.
Answers will vary.



Have students share their responses with a partner if time allows.